Multiscale Ship Detection Method in SAR Images Based on Information Compensation and Feature Enhancement

Mingming Zhu, Guoping Hu, Hao Zhou, and Shiqiang Wang

***Abstract*— With the development of synthetic aperture radar (SAR)-based imaging technology, SAR ship detection has achieved notable breakthroughs. Due to the scale diversity and class imbalance of ships, ship detection in SAR images is still a substantial challenge. To solve these problems, this article proposes a multiscale ship detection method for SAR images based on information compensation and feature enhancement. To improve the feature representations of multiscale ships, an information compensation module (ICM) and a feature enhancement module (FEM) are embedded into a feature pyramid network (FPN). Specifically, the ICM is constructed to extract and aggregate diverse spatial contextual information. At the same time, the FEM is introduced to solve the feature-level imbalance of the FPN by integrating multilevel ship features. In addition, a gradient density parameter is introduced to solve class imbalance problems. Experiments on a high-resolution SAR image dataset (HRSID) show that the proposed method achieves a comprehensive detection performance, i.e., 60.6% average precision (AP) and 67.1 frames/s (FPS), and outperforms other state-of-the-art methods.**

***Index Terms*— High-resolution synthetic aperture radar (SAR) images dataset (HRSID), multiscale, ship detection, SAR.**

I. Introduction

SYNTHEIC aperture radar (SAR) is an active microwave imaging sensor that can provide high-resolution SAR images in all weather and all day conditions [1]. Therefore, SAR is suitable for monitoring ships due to the changeable marine climate on Earth. Ship detection is conducive to traffic control, trade activity monitoring, fishery surveillance, and defense deployment. Although numerous methods, such as variance-weighted information entropy [2], the singular value decomposition network [3], automatic identification systems [4], the wavelet transform [5], the constant false alarm rate (CFAR) [6], and electromagnetic models [7], have been proposed, as a result of several difficult problems regarding the detection of multiscale ships in SAR images, this domain remains a hot topic. Most of the existing SAR ship detection methods use single-polarization amplitude images. On the physical side, these approaches may detect ships only if they have a strong return with respect to sea background. Therefore, considerable scattering interference caused by islands or the sea exists in SAR images.

To date, many traditional SAR ship detection methods have emerged. For example, Wang *et al.* [8] proposed an intensity-space domain CFAR ship detection method that makes good use of the intensity of each pixel and the correlations between pixels. However, this technique often depends on prior knowledge and background information. In [1], a modified CFAR approach based on object proposals is presented. The generator is used to create object proposals, and then, the proposal-based CFAR detector is used to detect the true positive (TP) object proposals. Therefore, this algorithm can achieve good detection performance in multiscale situations. Liang *et al.* [9] proposed an adaptive hierarchical detection method based on a coarse-to-fine mechanism, which includes a visual attention mechanism and a speed block kernel density estimation approach. The visual attention mechanism is used to strengthen ship targets. The speed block kernel density estimation method is used to obtain precise detection results. Wang *et al.* [10] proposed a fast multiscale patch-based contrast measure algorithm to detect ships in noisy SAR images. An improved sliding window structure and a target-based multiscale measurement strategy are used to reduce the noise sensitivity of the original algorithm and improve its detection speed. Lin *et al.* [11] proposed a superpixel-level Fisher vector for ship detection, but its shortcoming lies in its heavy time cost. Superpixels help to retain target outlines and suppress speckle noise. The superpixel-level Fisher vector better describes the discrimination between the target and clutter, further improving the detection performance of the model. The generalization abilities of traditional ship detection methods are relatively poor because it is difficult to design appropriate manual features to address different conditions. In addition, since a large number of parameters need to be defined in advance, the time costs of these methods are too high [12].

In recent years, convolutional neural networks (CNNs) have achieved remarkable success due to their powerful feature extraction and feature representation capabilities. CNN-based object detectors can be divided into two types. One includes two-stage detectors, such as the faster region-based

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CNN (Faster R-CNN) [13], which uses a region proposal network (RPN) to acquire a series of regions of interests (RoIs) and then performs object classification and localization based on the obtained proposal regions. The other is one-stage detectors, such as you only look once (YOLO) [14], the single-shot multibox detector (SSD) [15], and RetinaNet [16], which can transform the detection problem into a regression problem. Two-stage detectors can provide relatively good detection performance but have high time costs. In contrast, one-stage detectors are characterized by lower computational costs and higher real-time application value [12], [17].

The emergence of large-scale SAR datasets, such as the SAR ship detection dataset (SSDD) [18], high-resolution SAR images dataset (HRSID) [19], and large-scale SSDD-v1.0 (LS-SSDD-v1.0) [20], has made it possible to apply deep learning to detect objects in SAR images. First, CNNs were used in various ship detection tasks, such as RoI extraction and object recognition. At present, most ship detection algorithms adopt an end-to-end mode and directly output the obtained detection results, thereby improving their detection efficiency and accuracy [17]. For example, Lin *et al.* [21] proposed a new network named squeeze and excitation rank (SER) Faster R-CNN, which utilizes a multiscale feature map concatenation strategy to improve the quality of shared feature maps. An SER mechanism was applied to improve the performance of Faster R-CNN. The experimental results show that SER Faster R-CNN is a high-performance object detector. Although CNN-based methods have achieved good performance on the ImageNet dataset [22], the pattern analysis, statistical modeling and computational learning visual object classes (PASCAL VOC) dataset [23], and the Microsoft common objects in context (MS COCO) dataset [24], they do not perform well on SAR images [25]. The main reason for this is that SAR images provide a wider imaging coverage and larger imaging scale than images from ImageNet dataset, PASCAL VOC dataset, and MS COCO dataset. Moreover, some problems such as scene imbalance, densely distribution, scale diversity, and complex background are encountered in SAR ship detection.

To address the abovementioned problems in SAR ship detection, several CNN-based detectors have been proposed. Zhang *et al.* [26] proposed a balanced scene learning mechanism (BSLM) to solve the scene imbalance problem in SAR ship detection. The average precision (AP) of the BSLM was 92.22% on the SSDD. However, the BSLM does not consider the scale diversity of ships. Fu *et al.* [27] improved CenterNet to prevent it from missing detections. Although the AP of ship detection on the SAR-ship-dataset was 94.7%, the improved CenterNet still focused less on the scale diversity of ships. Pan *et al.* [28] proposed a multistage rotational region-based network for detecting arbitrary-oriented and densely arranged ships, and this approach achieved an AP of 90.11% on the SSDD. Zhao *et al.* [12] proposed an attention receptive pyramid network for detecting multiscale ships based on the SSDD; this approach achieved 98.2% and 84.1% AP values for offshore and inshore ships, respectively. Zhu *et al.* [29] proposed a high-speed lightweight detector named H2Det, which could achieve 92.0% AP on the HRSID. These ship detection methods [12], [28], [29] generally use feature pyramids to address the scale diversity of ships. However, some difficult problems remain with respect to detecting multiscale ships. One of these problems is that the feature pyramids used in SAR ship detection have weak feature representation abilities for multiscale ships. This is because the information at the highest level of a feature pyramid is lost due to feature channel reductions. In addition, the integrated features of a feature pyramid should contain balanced information from each resolution rather than focus solely on the adjacent resolution. The other problem is the imbalance between positive and negative samples and that between hard and easy samples during network training. Since most ships are sparsely distributed in SAR images, the number of positive samples is much larger than that the number of negative samples. In addition, we find that it is more difficult to accurately detect inshore ships than offshore ships. Therefore, class imbalance always exists in SAR ship detection tasks.

Therefore, improving the feature representation ability of the constructed feature pyramid and solving the class imbalance problem are essential for multiscale ship detection in SAR images. In this article, we present a novel multiscale ship detection method based on an information compensation module (ICM) and a feature enhancement module (FEM). RetinaNet is applied to the proposed model as the base network. The main contributions are as follows.

1) A novel object detector is proposed for multiscale ship detection in SAR images. Experimental results obtained on the HRSID demonstrate that the proposed method is better than other state-of-the-art methods.
2) To alleviate the information loss and single-scale contextual information issues of ships, the ICM is introduced to extract and aggregate diverse spatial contextual information.
3) To solve the feature-level imbalance of the constructed feature pyramid network (FPN), the FEM is introduced to integrate multilevel ship features.
4) To solve the class imbalance problem encountered in SAR ship detection, a gradient density parameter is introduced to reconstruct a new loss function.

The remainder of this article is organized as follows. The proposed method is described in Section II. Experimental results are presented and discussed in Section III, and the conclusion is drawn in Section IV.

II. Methodology

In this section, the details of the proposed method are introduced. First, the flowchart and framework of the proposed method are presented. Next, RetinaNet is briefly introduced as the baseline for the proposed method. Then, the ICM and FEM of the proposed method are described in detail. Finally, a new loss function is reconstructed.

A. Overall Flowchart and Framework

The overall flowchart of the proposed method, as shown in Fig. 1, consists of a backbone network, an improved feature pyramid, and classification and regression subnetworks.

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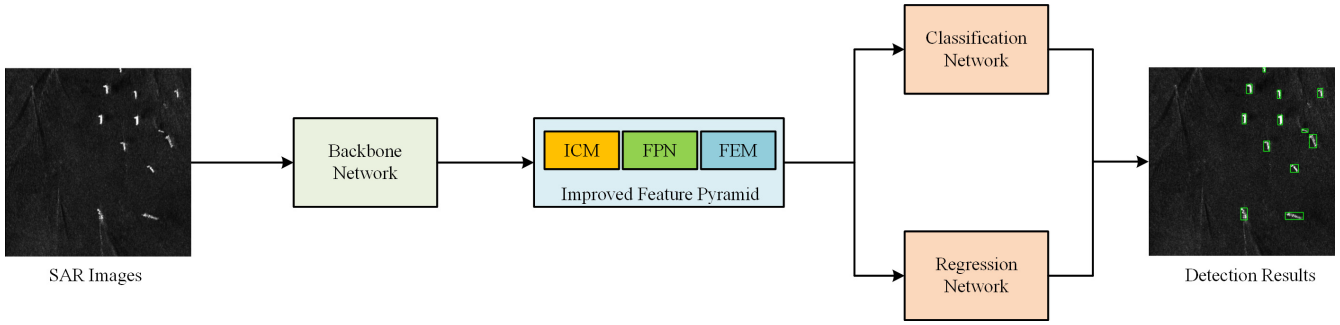


Fig. 1. Overall flowchart of the proposed method.

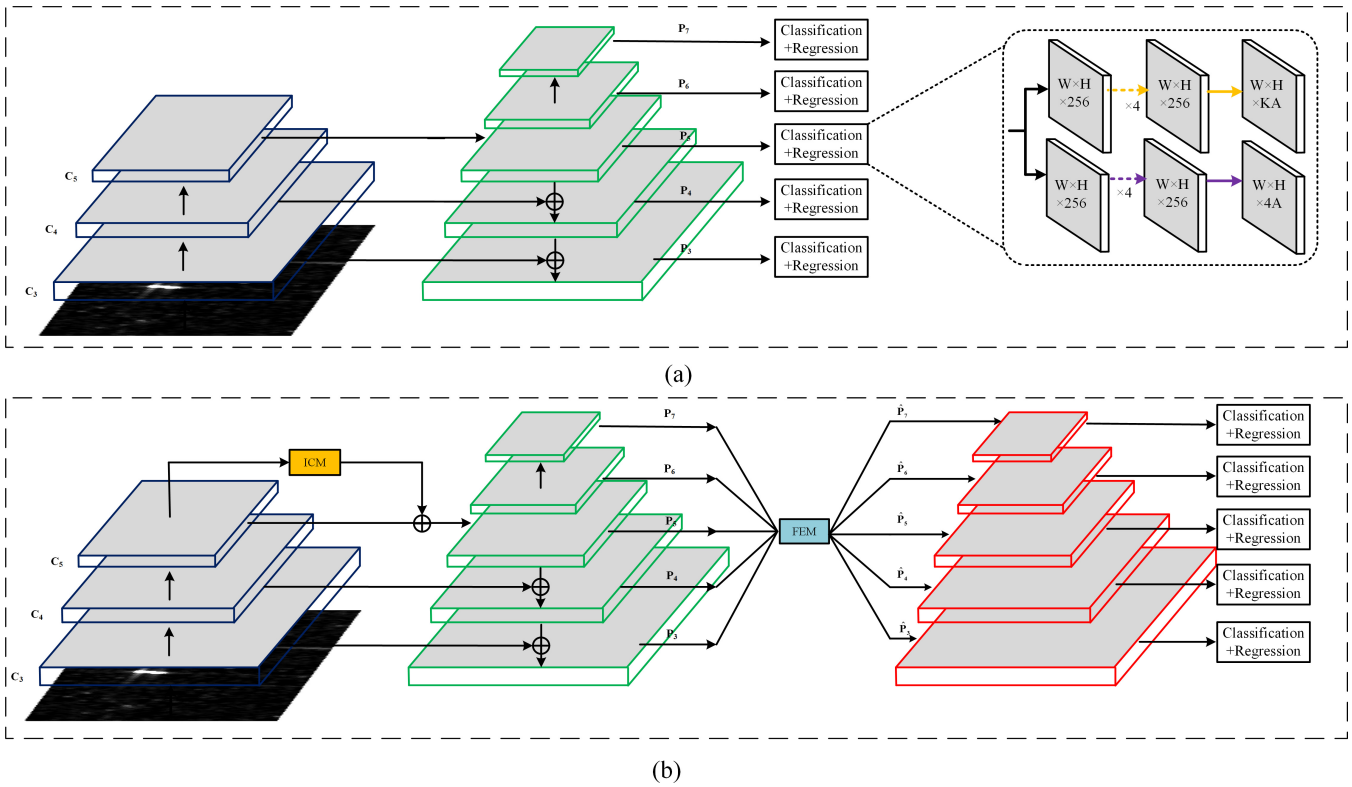


Fig. 2. Network architecture of RetinaNet and the proposed method. (a) RetinaNet. (b) Proposed method.

First, an SAR image is fed into the backbone network, which is utilized to extract basic feature maps through a bottom–up pathway. Second, an improved feature pyramid is established by using the ICM, FPN, and FEM in sequence. Ultimately, the final detection results are acquired by the classification and regression subnetworks.

As shown in Fig. 2(b), the overall framework of the proposed method builds on RetinaNet. Different from RetinaNet, the proposed method embeds the ICM and FEM into an FPN to construct an improved feature pyramid. In brief, the ICM is first introduced to extract and aggregate diverse spatial contextual information. Then, the FEM is introduced to integrate multilevel ship features. In addition, we use a reconstructed loss function to train the entire network in an end-to-end manner. Because RetinaNet is the baseline of the proposed method, the backbone network and FPN of the proposed method are basically the same as those of RetinaNet.

B. RetinaNet

As shown in Fig. 2(a), RetinaNet consists of an FPN [30] and two subnetworks (a classification subnet and a regression subnet). RetinaNet builds feature pyramid levels $\{P_3, P_4, P_5, P_6, P_7\}$ with $C = 256$ channels, where P_3–P_5 are computed from C_3 to C_5 via lateral connections and a top–down pathway. P_6 is obtained via a 3×3 convolution with a stride of 2 on C_5. To improve the large-object detection ability of the network, P_7 is computed by applying the rectified linear unit (ReLU) function followed by a 3×3 convolution with a stride of 2 on P_6. RetinaNet does not use the high-resolution pyramid level P_2 because of computational costs.

The anchors of RetinaNet have areas of 32^2–512^2 on pyramid levels P_3–P_7, respectively. At each pyramid level, three aspect ratios $\{1{:}2, 1{:}1, 2{:}1\}$ and three sizes $\{2^0, 2^{1/3}, 2^{2/3}\}$ are used for anchors. Each anchor is assigned a length-K one-hot classification vector, where K is the number of object classes, and a four vector for bounding box regression.

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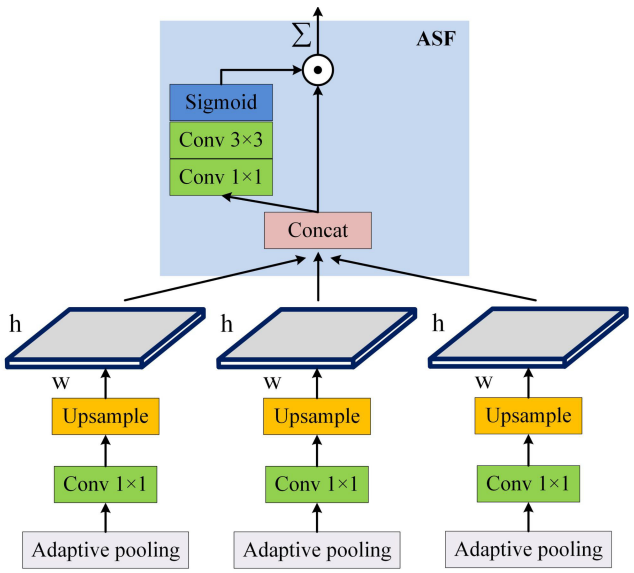


Fig. 3. ICM.

The classification subnet is attached to one of the pyramid levels, and the subnet and pyramid level share parameters with each other. Specifically, the classification subnet consists of four sets of $3 \times 3 \times 256$ convolutional layers followed by ReLU activations, a $3 \times 3 \times KA$ convolutional layer, and a sigmoid activation function. Given a $W \times H \times 256$ feature map, KA binary predictions are output per spatial location. In parallel with the classification subnet, the structure of the regression subnet is exactly the same as that of the classification subnet except for the output. For each spatial position in a case with A anchors, $4A$ outputs are used to predict the relative offsets. In addition, a focal loss is utilized to address the class imbalance problem.

C. Information Compensation Module

To improve the feature expression ability of the proposed method for SAR ship detection, the highest level feature map C_5 propagates along a top–down path and is gradually merged with lower level feature maps $\{C_3, C_4\}$. However, feature map C_5 suffers from information loss as a result of the reduction in the number of channels and only contains single-scale contextual information [31]. It is well-known that context information is important for object detection. Therefore, this article proposes the ICM to extract and aggregate diverse spatial context information, as shown in Fig. 3.

First, a scale-invariant adaptive average pooling is used to generate multiscale feature maps with diverse spatial context information

$$F_i = \text{AvgPool}(C_5, S \times \alpha_i), \quad i = 1, 2, 3 \tag{1}$$

where $S = w \times h$ represents the scale of feature map C_5 and $S \times \alpha_i$ represents the scale of F_i.

A 1×1 convolutional layer is used to reduce the number of channels to 256

$$F_i' = \text{Conv}_{1\times1}(F_i, 256), \quad i = 1, 2, 3. \tag{2}$$

The above features are upsampled through bilinear interpolation for the subsequent fusion operation

$$F_i'' = \text{Upsample}\left(F_i', S\right), \quad i = 1, 2, 3. \tag{3}$$

Considering the aliasing effect caused by interpolation, we build an adaptive spatial fusion module to fuse different features. Inspired by the squeeze-and-excitation block [32], the adaptive spatial fusion module generates a spatial weight map for each input feature

$$F'' = \text{Concat}\left(F_1'', F_2'', F_3''\right) \tag{4}$$

$$W = \text{Sigmoid}\left\{\text{Conv}_{3\times3}\left[\text{Conv}_{1\times1}\left(F''\right)\right]\right\}. \tag{5}$$

The spatial weight map W is used to obtain multiscale contextual information for feature map F'', and the feature map F_{ASF} is output

$$F_{\text{ASF}} = \sum_{i=1}^{3} \left(W_i \odot F_i''\right). \tag{6}$$

After summing F_{ASF} and C_5, a 3×3 convolutional layer is used to generate a feature map P_5 at the highest pyramid level

$$P_5 = \text{Conv}_{3\times3}\left[F_{\text{ASF}} + \text{Conv}_{1\times1}(C_5)\right]. \tag{7}$$

D. Feature Enhancement Module

Low-level feature maps provide high resolutions but limited semantic information and are thus suitable for small-scale ship detection. Comparatively, high-level feature maps provide more semantic information at low resolutions, making them suitable for large-scale ship detection [17]. Therefore, feature-level imbalances create challenges when performing multiscale SAR ship detection. FPN [30] was proposed with lateral connections to enrich the observed semantic information. After that, a path augmentation network [33] was proposed with a bottom–up pathway to further increase the low-level information contained in deep layers. However, the aforementioned methods make the integrated features focus more on adjacent resolutions but less on other resolutions. The semantic information contained in nonadjacent levels becomes diluted once per fusion step during the information flow process. This shows that the integrated features of the feature pyramid should contain balanced information from each resolution. Unlike previously methods that integrate multilevel features through lateral connections, the FEM is introduced to integrate the multilevel features of ship objects, as shown in Fig. 4.

First, we uniformly adjust the feature pyramid levels $\{P_3, P_4, P_5, P_6, P_7\}$ to the same resolution as that of P_5. P_3 and P_4 adopt maximum pooling operations, and P_5–P_7 adopt nearest interpolation operations

$$P_i' = \text{MaxPool}(P_i), \quad i = 3, 4 \tag{8}$$

$$P_i' = \text{NearestInterpolation}(P_i), \quad i = 5, 6, 7. \tag{9}$$

Next, balanced semantic features are obtained by averaging

$$\bar{P} = \frac{1}{5} \sum_{i=3}^{7} P_i'. \tag{10}$$

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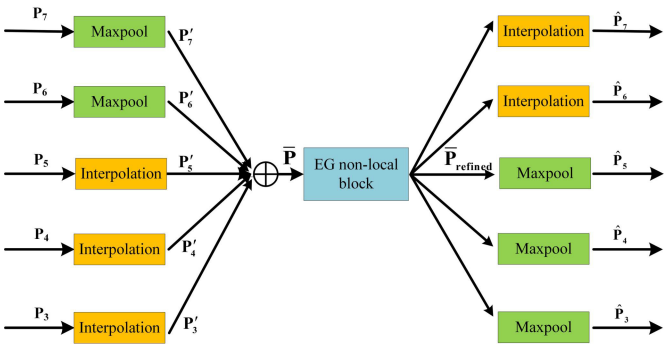


Fig. 4. FEM.

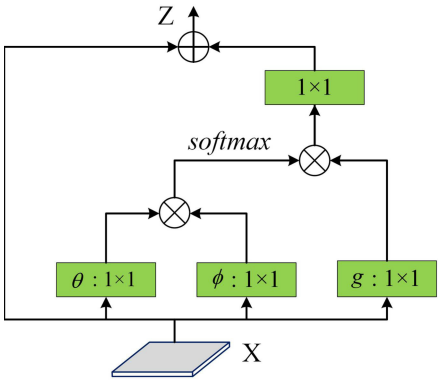


Fig. 5. Embedded Gaussian nonlocal block.

To further enhance the balanced semantic features, we use an embedded Gaussian nonlocal block [34] to achieve feature refinement, as shown in Fig. 5. The expression of the nonlocal operation is as follows:

$$y_i = \frac{1}{C(x)} \sum_j f\left(x_i, x_j\right) g\left(x_j\right) \tag{11}$$

where x is the input feature map, and y is the output feature map. f is a pairwise function used to calculate the correlation between x_i and x_j. g is a unary function, and $C(x)$ is a normalized function

$$g\left(x_j\right) = W_g x_j \tag{12}$$

where W_g is a weight matrix, which can be obtained by using a 1×1 convolutional layer.

The embedded Gaussian function [34] is defined as follows:

$$f\left(x_i, x_j\right) = e^{\theta(x_i)^{\mathrm{T}} \phi\left(x_j\right)} \tag{13}$$

$$\theta(x_i) = W_\theta x_i \tag{14}$$

$$\phi\left(x_j\right) = W_\phi x_j. \tag{15}$$

The normalized function $C(x)$ is defined as follows:

$$C(x) = \sum_j f\left(x_i, x_j\right). \tag{16}$$

By combining (11) and (16), we can obtain

$$y = \mathrm{softmax}\left(x^{\mathrm{T}} W_\theta^{\mathrm{T}} W_\phi x\right) g(x). \tag{17}$$

Therefore, the embedded Gaussian nonlocal block is defined as follows:

$$z = W_z y + x \tag{18}$$

where $+x$ denotes a residual connection.

After completing the above feature refinement steps, we can obtain a refined feature $\bar{P}_{\text{refined}}$. Next, we use maximum pooling and nearest interpolation to generate a new feature pyramid hierarchy

$$\hat{P}_i = \mathrm{NearestInterpolation}\left(\bar{P}_{\text{refined}}\right), i = 3, 4 \tag{19}$$

$$\hat{P}_i = \mathrm{MaxPool}\left(\bar{P}_{\text{refined}}\right), i = 5, 6, 7. \tag{20}$$

E. Loss Function Reconstruction

Since ships are generally sparsely distributed in SAR images, the number of positive samples is much larger than the number of negative samples. In addition, we find that it is more difficult to accurately detect inshore ships than offshore ships. Therefore, class imbalance generally exists in SAR ship detection. The focal loss [16] is a common loss function used by single-stage detectors to solve class imbalance problem. However, this function has certain limitations. On the one hand, focusing too much on particular difficult samples can lead to inappropriate results. On the other hand, the two hyperparameters in the focal loss function require artificial designs and joint tuning. In this article, we introduce a gradient density parameter to reconstruct the loss function.

Given a predicted probability $p \in [0, 1]$ and the corresponding ground-truth label $p^* \in \{0, 1\}$, the binary cross-entropy loss function is [35] defined as follows:

$$L_{\mathrm{CE}}\left(p, p^*\right) = \begin{cases} -\log(p), & p^* = 1 \\ -\log(1-p), & p^* = 0. \end{cases} \tag{21}$$

The model output x satisfies the sigmoid function

$$p = \frac{1}{1 + e^{-x}}. \tag{22}$$

The gradient with respect to x is defined as follows:

$$\frac{\partial L_{\mathrm{CE}}}{\partial p} = \begin{cases} -1/p, & p^* = 1 \\ 1/(1-p), & p^* = 0 \end{cases} \tag{23}$$

$$\frac{\partial p}{\partial x} = \frac{e^{-x}}{(1 + e^{-x})^2} = p(1-p) \tag{24}$$

$$\frac{\partial L_{\mathrm{CE}}}{\partial x} = \frac{\partial L_{\mathrm{CE}}}{\partial p} \frac{\partial p}{\partial x} = \begin{cases} p - 1, & p^* = 1 \\ p, & p^* = 0. \end{cases} \tag{25}$$

By using the above equation, g is defined as follows:

$$g = \left| \frac{\partial L_{\mathrm{CE}}}{\partial x} \right| == \begin{cases} 1 - p, & p^* = 1 \\ p, & p^* = 0. \end{cases} \tag{26}$$

The value of g represents the attribute (hard or easy) of a sample and reflects the sample's influence on the global gradient. Studies have shown that the number of very easy samples is very large, and these samples have a great impact on the global gradient. In addition, the number of very

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hard samples is greater than the number of moderately hard samples. The gradient directions of these very hard samples are largely different from those of other samples, which can lead to such samples being considered outliers.

The gradient density function is defined as follows:

$$\delta_\varepsilon(m,n)=\begin{cases}1, & n-\dfrac{\varepsilon}{2}\le m<n+\dfrac{\varepsilon}{2}\\ 0, & \text{otherwise}\end{cases} \tag{27}$$

$$l_\varepsilon(g)=\min\left(g+\frac{\varepsilon}{2},1\right)-\max\left(g-\frac{\varepsilon}{2},0\right) \tag{28}$$

$$\mathrm{GD}(g)=\frac{1}{l_\varepsilon(g)}\sum_{k=1}^{N}\delta_\varepsilon(g_k,g) \tag{29}$$

where $\sum_{k=1}^{N}\delta_\varepsilon(g_k,g)$ denotes the number of samples lying in the region centered at g with a length of ε. $l_\varepsilon(g)$ denotes the valid length of the region. N denotes the total number of samples.

The gradient density parameter is defined as follows:

$$\beta_i=\frac{N}{\mathrm{GD}(g_i)}. \tag{30}$$

Therefore, the reconstructed classification loss function is defined as follows:

$$L_C=\frac{1}{N}\sum_{i=1}^{N}\beta_i L_{\mathrm{CE}}\left(p_i,p_i^*\right)=\sum_{i=1}^{N}\frac{L_{\mathrm{CE}}\left(p_i,p_i^*\right)}{\mathrm{GD}(g_i)}. \tag{31}$$

The regression loss function is typically based on a smooth L_1 loss function [35]

$$L_{\mathrm{reg}}=\sum_{i\in\{x,y,w,h\}}\mathrm{SL}_1(t_i-v_i) \tag{32}$$

$$\mathrm{SL}_1(t_i-v_i)=\mathrm{SL}_1(z)=\begin{cases}0.5z^2, & |z|<1\\ |z|-0.5, & \text{otherwise}\end{cases} \tag{33}$$

where $t=(t_x,t_y,t_w,t_h)$ indicates the predicted offsets and $v=(v_x,v_y,v_w,v_h)$ denotes the target offsets.

To apply the gradient density parameter in the regression loss function, we use the ASL_1 loss [36] to replace the smooth L_1 loss function. The specific form of this approach is as follows:

$$\mathrm{ASL}_1(z)=\sqrt{z^2+\mu^2}-\mu \tag{34}$$

where $\mu=0.02$.

Given that $z=t_i-v_i$, the gradient of the ASL_1 loss with respect to t_i is

$$\begin{aligned}\frac{\partial \mathrm{ASL}_1}{\partial t_i}&=\frac{\partial \mathrm{ASL}_1}{\partial d}\frac{\partial d}{\partial t_i}\\ &=\frac{z}{\sqrt{z^2+\mu^2}}.\end{aligned} \tag{35}$$

g' is defined as follows:

$$g'=\left|\frac{z}{\sqrt{z^2+\mu^2}}\right|. \tag{36}$$

Therefore, the reconstructed regression loss function is defined as follows:

$$L_R=\frac{1}{N}\sum_{i=1}^{N}\beta_i\mathrm{ASL}_1(z_i)=\sum_{i=1}^{N}\frac{\mathrm{ASL}_1(z_i)}{\mathrm{GD}\left(g_i'\right)}. \tag{37}$$

III. Experiments

All experiments are implemented in the PyTorch framework with an Intel i9-9820X CPU and four GeForce RTX 2080Ti GPUs. We choose the MMDetection object detection toolbox for training and testing.

A. Dataset

In this article, the HRSID [19] is used to evaluate the detection performance of the proposed method. The HRSID, proposed by Wei *et al*. [19], was constructed by using original SAR images from the Sentinel-1B, TerraSAR-X, and TanDEM-X satellites. The HRSID contains 5604 images with 800 × 800 pixels and 16 951 multiscale ships. These images have various polarizations, imaging modes, and imaging conditions. The entire dataset is divided into a training set with 3642 images and a test set with 1962 images. MS COCO is the most challenging object detection dataset available today, which contains 91 common object categories with 82 of them having more than 5000 labeled instances. In total, the dataset has 2 500 000 labeled instances in 328 000 images. MS COCO is significantly larger in number of instances per category than the ImageNet dataset and PASCAL VOC dataset, which may aid in learning contextual information. MS COCO contains considerably more object instances per image (7.7) as compared to ImageNet (3.0) and PASCAL VOC (2.3). Furthermore, MS COCO dataset contains 3.5 categories per image as compared to PASCAL (1.4) and ImageNet (1.7) on average. The biggest class is the person, nearly 800 000 instances, while the smallest class is hair driver, about 600 instances in the whole dataset. Except for 20 classes with many or few instances, the number of instances in the remaining 71 categories is roughly the same. We use the definition standards from the MS COCO dataset regulation to determine small ships, medium ships, and large ships. As a result, the ships with their bounding box areas below 32 × 32 pixels are regarded as small ones, area from 32 × 32 pixels to 96 × 96 pixels as medium ones, and area above 96 × 96 pixels as large ones. The numbers of small ships, medium ships, and large ships are 9242 (54.5%), 7388 (43.5%), and 321 (2%), respectively. Some samples and shape distributions derived from the HRSID are shown in Fig. 6(a) and (b), respectively.

B. Parameter Settings

In all experiments, ResNet-50 is used as the backbone network of the proposed method. The anchors of the proposed method have the areas of 32^2–512^2 at pyramid levels P_3–P_7, respectively. At each pyramid level, the sizes and aspect ratios of the anchors are set to {0.5, 1, 2} and $\{2^0, 2^{1/3}, 2^{2/3}\}$, respectively. The SAR images of the proposed method are proportionally resized to 800 × 800 pixels during the training and testing processes. α_1, α_2, and α_3 in (1) are set as 0.1, 0.2, and 0.3, respectively. ε in (27) is set as 0.1. The proposed method is trained for 24 epochs. The initial learning rate is 0.01 and decays 10 times after 16 epochs and 22 epochs, as shown in Fig. 7(a). Fig. 7(b)–(d) shows the classification

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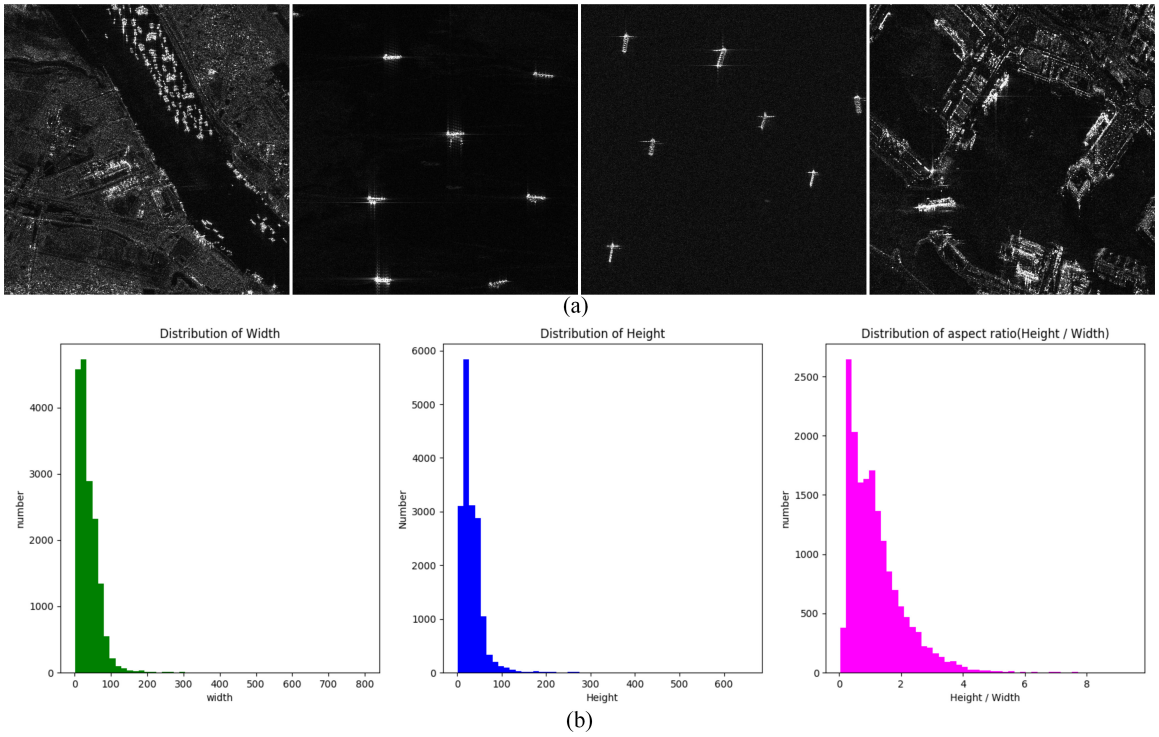


Fig. 6. Some samples and shape distributions of HRSID. Width and Height represent the width and height of the ground truth box, respectively. (a) Samples. (b) Shape distributions.

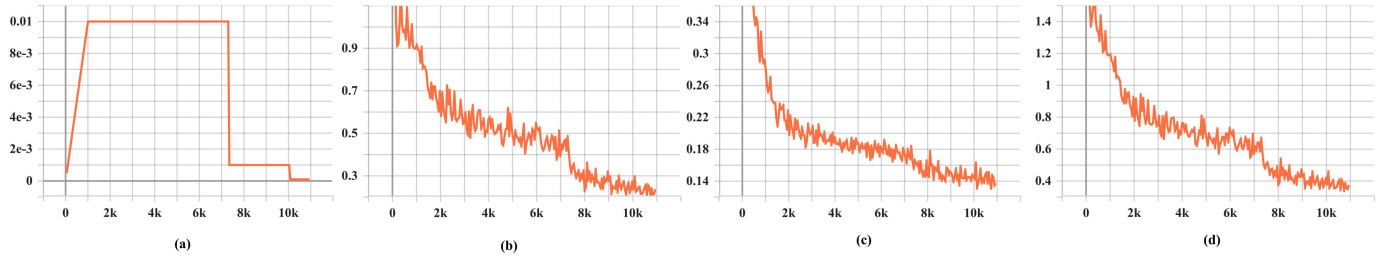


Fig. 7. Training information. (a) Learning rate. (b) Classification loss. (c) Regression loss. (d) Total loss.

loss, regression loss, and total loss of the proposed method, respectively.

In the comparison experiment, ResNet-50 is used as the backbone network of the Faster RCNN [13], NAS-FPN [37], Libra RCNN [38], and PAFPN [33]. VGG-16 is used as the backbone network of the SSD [15]. Darknet-53 is used as the backbone network of YOLOv3 [39]. The SAR images of the Faster RCNN, Libra RCNN, and PAFPN are proportionally resized to 800 × 800 pixels during the processes of training and testing. The SAR images of the NAS-FPN are proportionally resized to 640 × 640 pixels during the processes of training and testing. The SAR images of the SSD are proportionally resized to 300 × 300 pixels. The SAR images of YOLO are proportionally resized to 416 × 416 pixels. The Faster RCNN, SSD, Libra RCNN, and PAFPN are trained for 24 epochs. The NAS-FPN is trained for 50 epochs. YOLO is trained for 273 epochs. For the Faster RCNN, NAS-FPN, Libra RCNN, and PAFPN, the initial learning rate is 0.01 with a decay ratio of 0.1. The initial learning rate of the SSD is 0.002, and the decay ratio is 0.1. The initial learning rate of YOLO is 0.001 with a decay ratio of 0.1. The other hyperparameters in MMDetection are set to their default values.

C. Evaluation Metrics

If the intersection over union (IoU) between a bounding box and its corresponding ground truth is higher than the preset IoU threshold, it means that the associated ship is detected correctly. The numbers of TP samples, false positive (FP) samples, and false negative (FN) samples are used to define the precision, recall, and AP [13]

$$\text{Precision} = \frac{\text{TP}}{\text{TP+FP}} \tag{38}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP+FN}} \tag{39}$$

$$\text{AP} = \int_0^1 P(R)dR \times 100\% \tag{40}$$

where $P(R)$ is the precision-recall curve.

In this article, we adopt AP, AP_{50}, AP_{75}, AP_{S}, AP_{M}, and AP_{L} as evaluation metrics. AP_{50} and AP_{75} denote IoU

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TABLE I
DETECTION RESULTS OF ABLATION STUDIES

	AP (%)	AP_{50} (%)	AP_{75} (%)	AP_S (%)	AP_M (%)	AP_L (%)	FPS	Parameter (M)	FLOPs (G)
RetinaNet [17]	58.2	83.0	64.4	59.2	62.4	25.0	68.8	36.1	117.6
RetinaNet +ICM	60.3	84.3	67.2	**61.5**	63.6	31.7	67.7	37.9	117.8
RetinaNet +FEM	60.1	84.4	66.9	61.2	64.4	35.6	68.1	36.3	118.2
RetinaNet +GD	59.2	84.2	66.2	59.8	64.3	**37.4**	68.7	36.1	117.6
The proposed method	**60.6**	**85.5**	**67.9**	61.4	**65.5**	32.0	67.1	38.2	118.4

thresholds of 0.5 and 0.75, respectively. AP denotes the average AP yielded under different IoU thresholds from 0.5 to 0.95 with a step of 0.05. AP_S, AP_M, and $AP_L$are adopted to evaluate the detection performance of the model for small, medium, and large ships, respectively. We use parameter to represent parameter quantity and use floating point operations (FLOPs) to represent the computational cost. In addition, we use frames/s (FPS) [13] to represent the detection speed, and this measure is defined as follows:

$$\text{FPS}=\frac{1}{\text{Time}}. \tag{41}$$

D. Ablation Study

To verify the effectiveness of the proposed modules, we perform the following ablation studies. Since the proposed method is similar in structure to RetinaNet, RetinaNet is used as the baseline detector. Table I shows the detection results of RetinaNet, RetinaNet with the ICM (RetinaNet + ICM), RetinaNet with the FEM (RetinaNet + FEM), RetinaNet with the gradient density parameter (RetinaNet + GD), and RetinaNet with the ICM, FEM, and gradient density parameter (the proposed method).

1) Ablation Study on the ICM: It can be seen from Table I that the ICM can obviously improve the detection performance of the proposed method. Compared with the baseline detector, the ICM increases the AP, AP_{50}, AP_{75}, AP_S, AP_M, and AP_L by 2.1%, 1.3%, 2.8%, 2.3%, 1.2%, and 6.7%, respectively. The effectiveness of the ICM can also be verified in Fig. 8. As shown in Fig. 8(b), more false alarms are produced by the baseline detector. In contrast, RetinaNet + ICM has fewer false alarms. These comparison results demonstrate the effect of the ICM, which can yield improved detection performance by reducing the number of false alarms. Due to the limitations of the FPN, information loss and single-scale contextual information issues exist in RetinaNet, resulting in some false alarms. However, the ICM extracts and aggregates the diverse spatial context information of ships, which can reduce the information loss and improve the feature pyramid simultaneously. Therefore, the detection performance of RetinaNet + ICM is superior to that of RetinaNet. Although the computational cost (FLOPs) of RetinaNet + ICM increases (117.8G > 117.6G), it is acceptable because RetinaNet + ICM trades 0.17% [(117.8 − 117.6)/117.6 = 0.0017] FLOPs for a 3.61% [(60.3 − 58.2)/58.2 = 0.0361] AP increase.

2) Ablation Study on the FEM: As shown in Table I, the AP, AP_{50}, AP_{75}, AP_S, AP_M, and AP_L of RetinaNet + FEM are 1.9%, 1.4%, 2.5%, 2.0%, 2.0%, and 10.6% higher than those that RetinaNet, respectively. These results suggest that the FEM is effective for improving detection accuracy. The effectiveness of the FEM can also be verified in Fig. 9. It can be seen that RetinaNet has more false alarms. In contrast, RetinaNet + FEM has fewer false alarms. These comparison results demonstrate the effect of the FEM, which can improve detection performance by reducing the number of false alarms. This is because RetinaNet integrates multilevel features by using lateral connections, which makes the integrated features focus more on adjacent resolutions but less on other resolutions. The semantic information contained in nonadjacent levels is diluted once per fusion during the information flow. However, the FEM enhances the original features by integrating balanced semantic features. In this way, each resolution in the pyramid obtains equal information from the other resolutions, thus balancing the information flow and causing the features to be more discriminative. Although the computational cost (FLOPs) of RetinaNet + ICM increases (118.2G > 117.6G), it is very small. RetinaNet + ICM trades 0.51% [(118.2 − 117.6)/117.6 = 0.0051] FLOPs for a 3.26% [(60.1 − 58.2)/58.2 = 0.0326] AP increase.

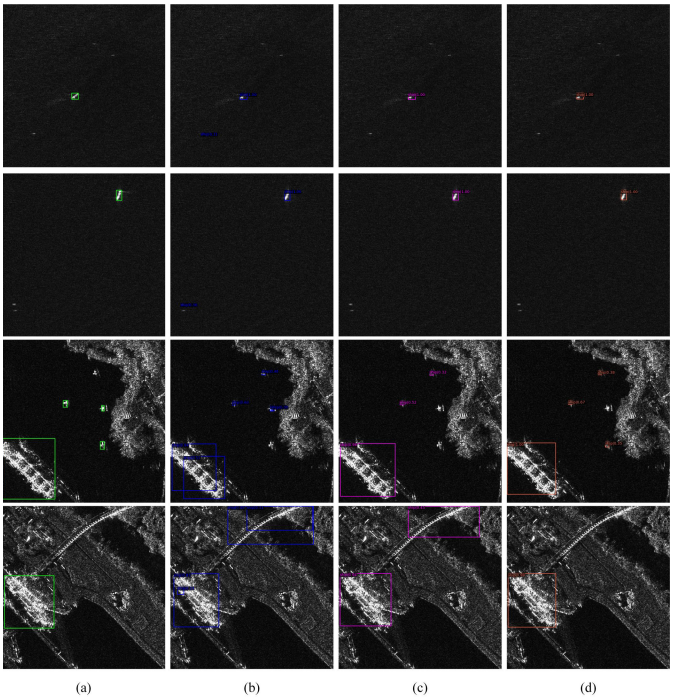


Fig. 8. Comparison results of RetinaNet, RetinaNet + ICM, and the proposed method. (a) Ground truth. (b) RetinaNet. (c) RetinaNet + ICM. (d) Proposed method.

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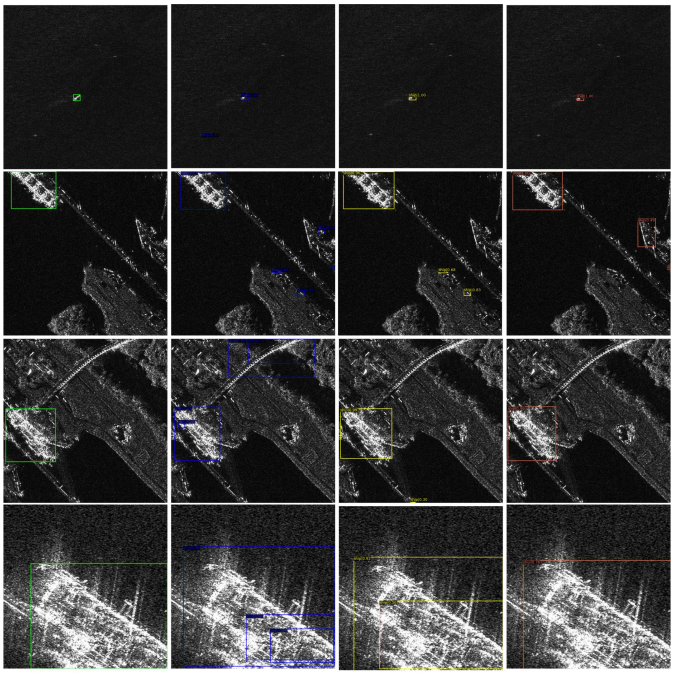


Fig. 9. Comparison results of RetinaNet, RetinaNet + FEM, and the proposed method. (a) Ground truth. (b) RetinaNet. (c) RetinaNet + FEM. (d) Proposed method.

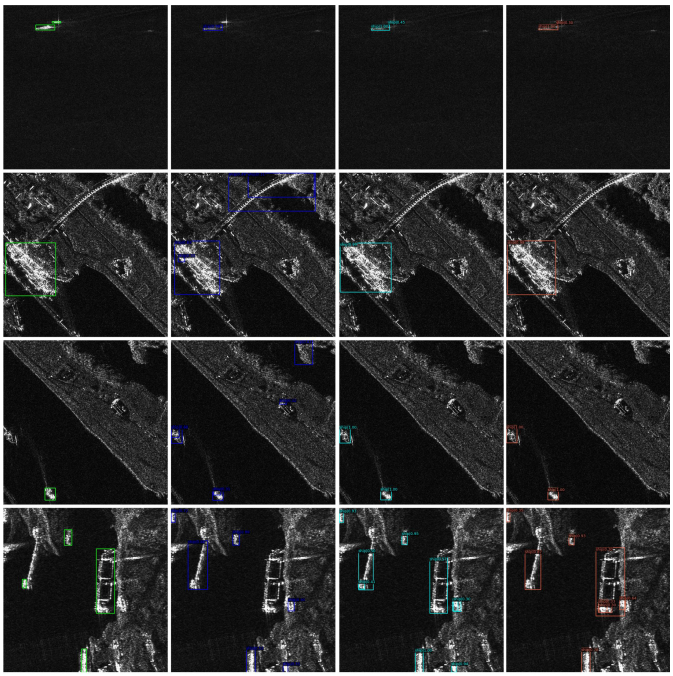


Fig. 10. Comparison results of RetinaNet, RetinaNet + GD, and the proposed method. (a) Ground truth. (b) RetinaNet. (c) RetinaNet + GD. (d) Proposed method.

3) Ablation Study on the Gradient Density Parameter: As shown in Table I, AP, AP_{50}, AP_{75}, AP_S, AP_M, and AP_L of RetinaNet + GD are 1.0%, 1.2%, 1.8%, 0.6%, 1.9%, and 12.4% higher than those of RetinaNet, respectively. These results suggest that the GD parameter can obviously improve the detection performance of the proposed method, especially in terms of the AP_L index. The effectiveness of the GD parameter can also be verified in Fig. 10. As shown in Fig. 10(b), RetinaNet has more false alarms with several missed detection results. In contrast, RetinaNet+GD has fewer false alarms and missed ships. These comparison results demonstrate the effect of the GD parameter, which can improve detection performance by reducing the numbers of false alarms and missed ships. This is because the focal loss of RetinaNet is a static loss, which cannot be adopted to change the data distribution. In addition, the focal loss focuses too much on particularly difficult samples, resulting in inappropriate model results. However, the GD parameter is used for dealing with class imbalance in the proposed method. Therefore, the training process can be more efficient and stable. The introduction of the GD does not bring an additional computational cost (FLOPs).

4) Ablation Study on the ICM, FEM, and GD Parameter: As shown in Table I, AP, AP_{50}, AP_{75}, AP_S, AP_M, and AP_L of the proposed method are 2.4%, 2.5%, 3.5%, 2.2%, 3.1%, and 7.0% higher than those of RetinaNet, respectively. Although the AP_S and AP_L of the proposed method are not the highest, the proposed method can obtain a better comprehensive detection performance. As shown in Figs. 8(d), 9(d), and 10(d), the proposed method has fewer false alarms and missed detections, that is, it has the best detection performance in comparison with other methods. The main reason for this is that the advantages of the three modules are combined in the proposed method. In addition, the computational cost (FLOPs) of the proposed method is slightly higher than that of RetinaNet (118.4G > 117.6G). However, the AP of the proposed method is 1.04 (60.6/58.2 = 1.04) times higher than that of RetinaNet. Therefore, the proposed method is still excellent because the proposed method trades 0.68% [(118.4 − 117.6)/117.6 = 0.0068] FLOPs for a 4.12% [(60.6 − 58.2)/58.2 = 0.0412] AP increase.

E. Detection Results

Fig. 11 shows the ship detection results of the proposed method on the HRSID dataset. It is clear that the proposed method almost detects all the ships on the ocean, including ships in offshore or inshore areas. The result confirms that our method is effective for detecting the multiscale ships in practice. Although our method achieves excellent performance, a few missing ships and false alarms still exist. Some missing ships present very weak or low intensity, so that they would induce few responses on the shallow layers, in turn leading to them being missed. Besides these missing targets, some false alarms are also observed. They mainly come from some building facilities on land, some harbor facilities in the open ocean area, or near the coast, which are similar to ships in shape and intensity.

F. Comparison With Other State-of-the-Art Methods

To demonstrate the superiority of the proposed method, we compare it with other state-of-the-art methods, such as the

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TABLE II
COMPARISON WITH OTHER STATE-OF-THE-ART METHODS

Methods	Input sizes	AP (%)	AP_{50} (%)	AP_{75} (%)	AP_S (%)	AP_M (%)	AP_L (%)	FPS	Parameter (M)	FLOPs (G)
SSD [16]	300×300	47.0	74.4	52.3	47.0	62.2	26.0	**101.4**	**23.8**	34.3
NAS-FPN [37]	640×640	51.7	76.3	56.5	51.0	65.2	21.7	59.9	58.1	124.8
YOLOv3 [39]	416×416	53.6	**86.2**	57.3	53.4	60.6	26.6	88.2	61.5	**32.8**
Faster RCNN [14]	800×800	53.7	76.7	59.9	53.9	60.4	12.9	23.3	32.8	792.8
PAFPN [33]	800×800	59.3	79.7	**68.1**	59.9	**65.7**	30.2	66.1	44.7	139.2
Libra RCNN [38]	800×800	59.5	78.8	67.9	59.9	66.6	23.6	66.3	41.4	125.5
The proposed method	800×800	**60.6**	85.5	67.9	**61.4**	65.5	**32.0**	67.1	38.2	118.4

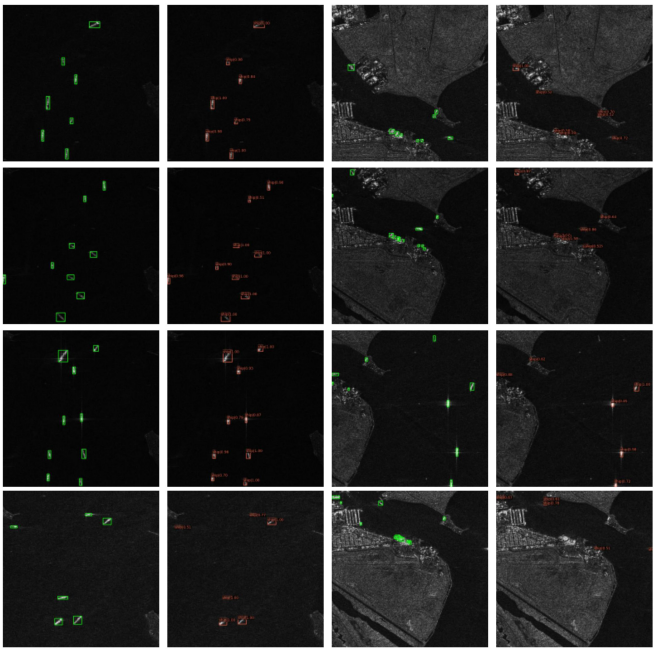


Fig. 11. Detection Results of the proposed method on the HRSID dataset.

SSD [15], NAS-FPN [37], YOLOv3 [39], Faster RCNN [13], PAFPN [33], and Libra RCNN [38], which are implemented by the MMDetection object detection toolbox. The AP, AP_{50}, AP_{75}, AP_S, AP_M, AP_L, parameter, FLOPs, and FPS are adopted as the evaluation metrics of the different methods on the HRSID. As illustrated in Table II, we can draw the following conclusions.

1) The AP value of the proposed method is 60.6%, which is 13.6%, 8.9%, 7%, 6.9%, 1.3%, and 1.1% higher than those of the SSD, NAS-FPN, YOLOv3, Faster RCNN, PAFPN, and Libra RCNN, respectively. Because the SSD lacks a feature pyramid, it performs worse than the other methods. Therefore, the feature pyramid is important for multiscale object detection. The AP of the proposed method is higher than those of other methods with feature pyramids. This might be because of the ICM, FEM, and GD parameters introduced into the proposed method.
2) The AP_{50} value of YOLOv3 is the highest, and it is 0.7% higher than that of the proposed method. This might be because of the well-designed feature fusion and anchor generation strategies of YOLOv3. However, the AP and AP_{75} of the proposed method are 7% and 10.6% higher than those of YOLOv3, respectively. These results show that the localization of the proposed method is more precise than that of YOLOv3. The AP_{75} and AP_M of PAFPN are the highest. It might be because PAFPN enhances the feature maps in lower layers through bottom–up path augmentation. However, the AP of the proposed method is 1.3% higher than that of PAFPN. This result shows that the localization of the proposed method is more precise than that of PAFPN.
3) The AP_S, AP_M, and AP_L are adopted to evaluate the detection performance of the proposed method for small, medium, and large ships, respectively. The AP_S, AP_M, and AP_L of the proposed method are close to those of PAFPN and Libra RCNN. This is due to the contributions of the improved feature pyramids in PAFPN and Libra RCNN. Although the AP_M of the proposed method is not the highest, the proposed method can obtain a better comprehensive detection performance. This shows that the proposed method can achieve a better balance when conducting multiscale object detection.
4) The detection speeds (FPS) of SSD and YOLOv3 are obviously faster than those of other methods. This might be because of their lower computational costs (FLOPs) and smaller input image sizes. Since Faster RCNN is a two-stage detector, its detection speed (FPS) is the slowest. Since PAFPN, Libra RCNN, and the proposed method have similar structures (i.e., feature pyramids and the same input image sizes), their detection speeds (FPS) are very close.

To visually demonstrate the detection performance of different methods, Fig. 12 shows a comparison of the results produced by different methods on the HRSID. In the first column of Fig. 12, several ships are missed by all methods except Libra RCNN and PAFPN. However, Libra RCNN and PAFPN yield many false alarms. In addition, several false alarms are obtained with Faster RCNN, RetinaNet, and YOLOv3. Although one false alarm is produced with the proposed method, the associated probability is lower than those of other methods. In the second column of Fig. 12, some false alarms are yielded by all methods. However, the proposed method has fewer false alarms and the corresponding probabilities are lower. In the third column of Fig. 12, several false alarms are obtained by Faster RCNN, Libra RCNN, PAFPN, and RetinaNet. Although SSD, NAS-FPN, YOLOv3, and the proposed method do not produce false alarms, the proposed method

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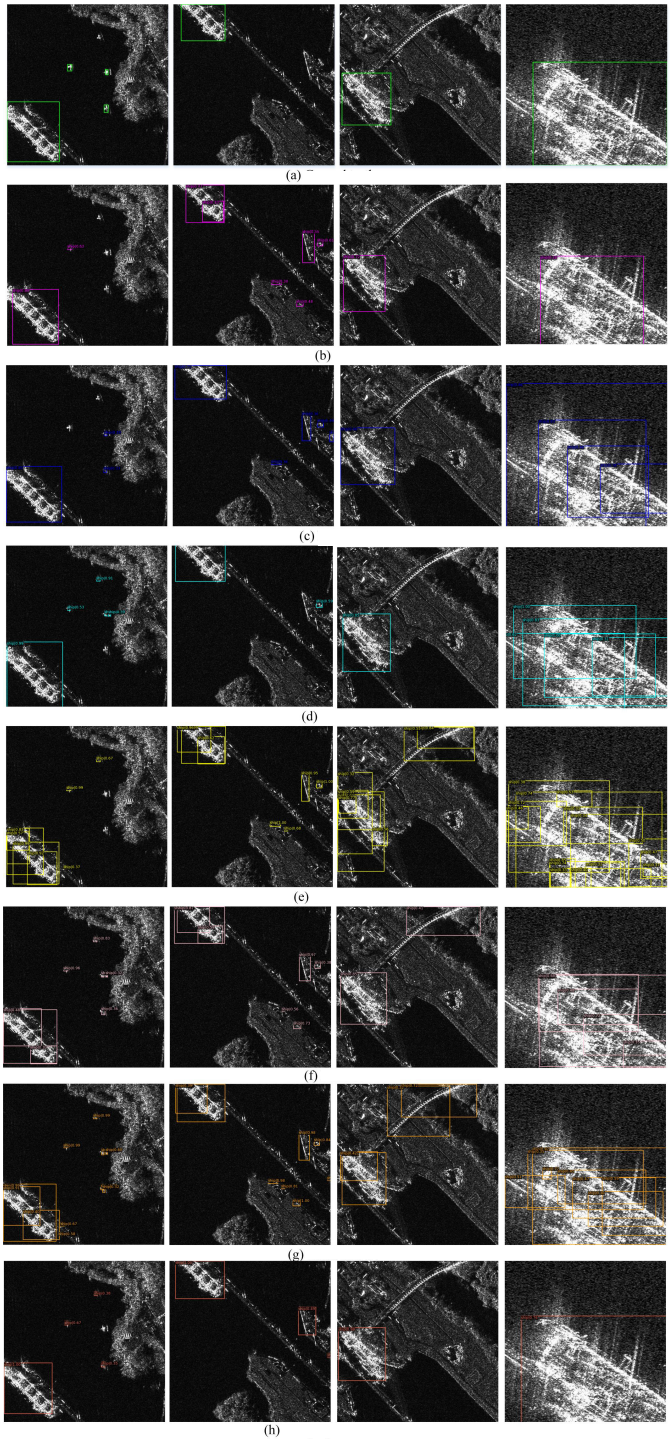


Fig. 12. Comparison with other state-of-the-art methods. (a) Ground truth. (b) SSD. (c) NAS-FPN. (d) YOLOv3. (e) Faster RCNN. (f) PAFPN. (g) Libra RCNN. (h) The proposed method.

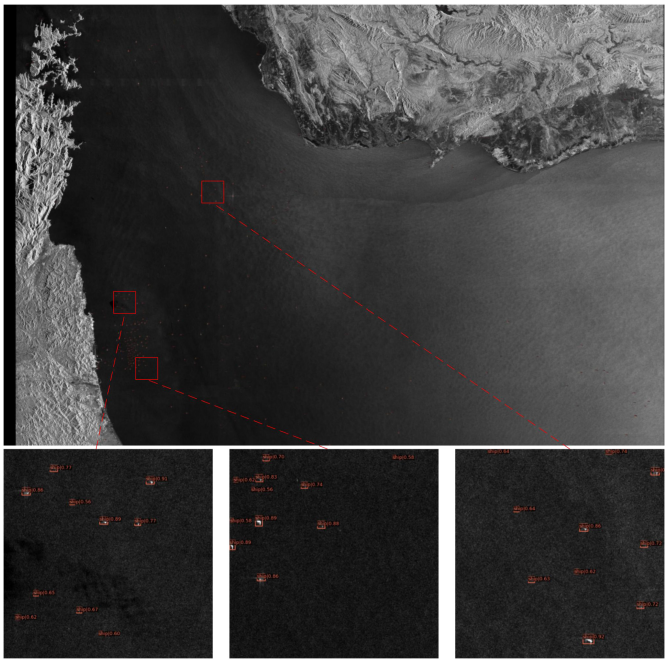


Fig. 13. Visual detection results of the proposed method on a large-scene Sentinel-1 SAR image.

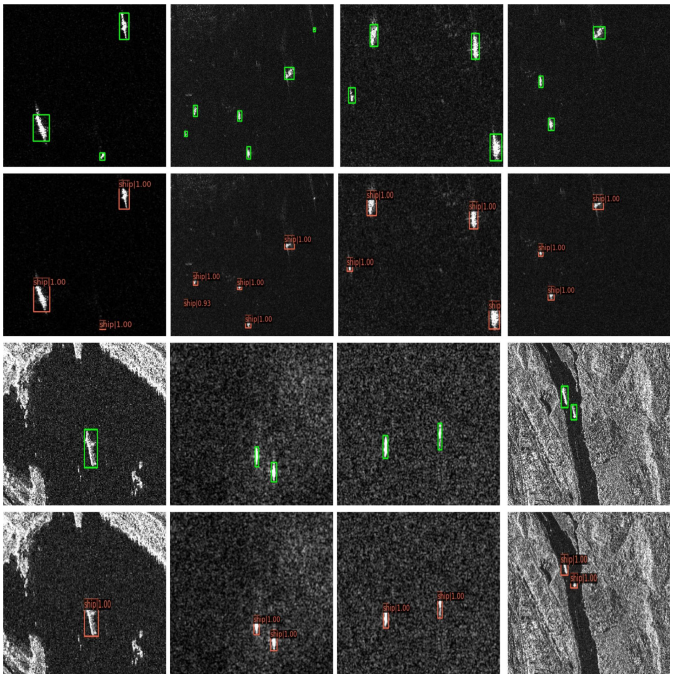


Fig. 14. Visual detection results of the proposed method on SSDD dataset. The green rectangles and red rectangles are correct ships and predicted ships, respectively.

can detect ships with higher probability and more accurate localization. In the fourth column of Fig. 12, false alarms exist for all methods except SSD and the proposed method. However, our probability is higher, and the localization of the proposed method is more accurate. The proposed method benefits from the feature pyramid improved by the ICM and the FEM, as well as the new loss function improved by the GD parameter, which can effectively improve the detection performance of the proposed method for multiscale ship detection. Overall, the proposed method is superior to other state-of-the-art methods.

G. Comparison With a CFAR Detector

To compare the performance of the proposed method with that of a classical two-parameter CFAR detector [40], we perform actual ship detection with a large-scene Sentinel-1 SAR image derived from LS-SSDD-v1.0. Table III shows its

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TABLE III
DETAILED DESCRIPTIONS OF THE LARGE-SCENE SENTINEL-1 SAR IMAGE

Place	Date	Time	Polarization	Mode	Resolution	Image Size (Pixels)
Gulf of Cadiz	18 June 2020	09:15:04	VV	IW	5 m ×20 m	26,000 × 16,000

VV denotes the co-polarization channel. IW denotes the interferometric wide swath.

TABLE IV
COMPARISON DETECTION RESULTS ON SENTINEL-1 SAR IMAGE

	GT	Detections	TP	FP	FN	Recall (%)	Precision (%)	F1	Time (s)
CFAR	351	556	314	242	37	89.5	56.5	0.69	735
The proposed method	351	338	306	32	45	87.2	90.5	0.89	152

TABLE V
DETECTION RESULTS OF THE PROPOSED METHOD ON SSDD DATASET

	AP (%)	AP_{50} (%)	AP_{75} (%)	AP_S (%)	AP_M (%)	AP_L (%)
The proposed method	61.3	96.6	72.9	59.3	66.5	45.3

descriptions. The traditional CFAR detector typically uses $F1$ score to measure accuracy, this metric is defined by

$$\mathrm{F1} = 2 * \frac{\mathrm{Precision} * \mathrm{Recall}}{\mathrm{Precision+Recall}}. \tag{42}$$

As shown in Table IV, the proposed method is greatly superior to the CFAR detector in terms of the $F1$ score ($0.89 > 0.69$) and time ($152 < 735$). Although the CFAR detector achieves a higher recall rate, its precision rate is rather low. Fig. 13 shows the visual detection results. These results demonstrate that the efforts needed to build a CNN are valuable.

H. Migration Ability

To verify the model migration ability of the proposed method, we perform an experiment on the SSDD [10]. We divide the entire dataset into two parts: a training set (1044) and a test set (116). Table V and Fig. 14 show the detection results obtained on the SSDD. The proposed method can achieve the state-of-the-art results, with AP, AP50, AP75, APS, APM, and APL values of 61.3%, 96.6%, 72.9%, 59.3%, 66.5%, and 45.3%, respectively. These results show that the proposed method displays strong migration ability. Fig. 14 shows that the proposed method can accurately detect most ship objects.

IV. CONCLUSION

In this article, a novel method based on RetinaNet is proposed for detecting multiscale ships in SAR images. First, it is known from this article that spatial information in feature maps is beneficial to detect small-scale ships, while semantic information is suitable for large-scale ships. Therefore, the proposed method adopts a pyramid structure for multiscale ship detection. Besides, to reduce the information loss, the diverse spatial context information is extracted and aggregated. Next, to solve the feature-level imbalance, the multilevel ship features are integrated by FEM rather than lateral connections. Finally, to solve class imbalance, the loss function is reconstructed by reducing the number of easy and difficult samples.

Experimental results on HRSID show that the proposed method achieves competitive detection performance in comparison with the SSD, NAS-FPN, YOLOv3, Faster RCNN, PAFPN, and Libra RCNN. Experimental results on a large-scene Sentinel-1 SAR image show that the proposed method is vastly superior to the CFAR detector. In addition, we verify the model migration ability of the proposed method on the SSDD. However, it should be noted that although the proposed method has better detection performance, it cannot completely eliminate all false alarms and missed detection results, which requires further analysis and research. The research on how to integrate polarization information into deep networks for ship detection in SAR images will be a key direction of the further research in the future.

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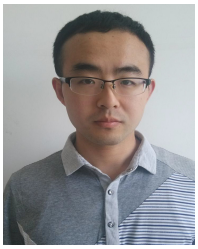
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